Chapter 3: Number Play

Worksheet 3: Divisibility Rules and Factors

Learning Objective: Students will apply divisibility rules for 2, 3, 4, 5, 6, 8, 9, 10, and 11; find factors, multiples, HCF, and LCM; identify prime and composite numbers.

Worked Example

Test whether 2,376 is divisible by 2, 3, 4, 6, and 9.
Solution:

- By 2: Last digit is 6 (even) $\Rightarrow$ **Yes**
- By 3: $2 + 3 + 7 + 6 = 18$, divisible by 3 $\Rightarrow$ **Yes**
- By 4: Last two digits 76; $76 \div 4 = 19$ $\Rightarrow$ **Yes**
- By 6: Divisible by both 2 and 3 $\Rightarrow$ **Yes**
- By 9: $2 + 3 + 7 + 6 = 18$, divisible by 9 $\Rightarrow$ **Yes**

Questions 1–5: Easy / Recall

Q1. State the divisibility rule for 5.

Q2. Is 4,320 divisible by 10? Why?

Q3. Is 531 divisible by 3? Show your working.

Q4. What is the smallest prime number greater than 10?

Q5. List all factors of 48.

Questions 6–10: Basic Application

Q6. Check 3,672 for divisibility by 2, 3, 4, 6, and 9.

Q7. Find the HCF of 36 and 84 using prime factorisation.

Q8. Find the LCM of 12, 15, and 20.

Q9. A number is divisible by both 4 and 6. Must it be divisible by 24? Give a counter-example.

Q10. Find all prime numbers between 50 and 80.

Questions 11–15: Practice and Skill Building

Q11. Find the HCF and LCM of 56 and 98. Verify: HCF $\times$ LCM $= 56 \times 98$.

Q12. The divisibility rule for 11: alternate digit sum difference. Test 54,978.

Q13. Find the smallest number divisible by all of 2, 3, 4, 5, 6.

Q14. Two numbers have HCF = 12 and LCM = 360. One number is 60. Find the other.

Q15. Express 120 as a product of prime factors.

Questions 16–18: Higher Order Thinking

Q16. Find the largest 5-digit number divisible by 36.

Q17. Three bells ring at intervals of 12, 18, and 24 minutes. They ring together at 8:00 AM. When will they next ring together? How many times do they all ring together between 8:00 AM and 12:00 PM?

Q18. A number N leaves remainder 3 when divided by 5, remainder 3 when divided by 7, and remainder 3 when divided by 9. Find the smallest such N. (Hint: $N - 3$ is divisible by LCM of 5, 7, 9.)

Questions 19–20: Real-Life Applications

Q19. A school has 252 boys and 336 girls. They must be seated in equal rows with either only boys or only girls in each row. Find the largest possible row size. How many rows of boys and girls are there?

Q20. At a mela, stalls are placed every 6 m, 8 m, and 12 m along three different roads from a common starting point. At what distance from the start will stalls from all three roads be at the same point for the first time?

Reflection Question: Why is 1 neither prime nor composite? How does this matter for prime factorisation?